

“AI Powered Voice-Guided Presentation: Revolutionizing PPT Explanations”

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Partial Fulfillment of the Requirements for the
8th Semester B.Tech. Project*

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Degree for which the Thesis is submitted: **Bachelor of Technology**

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It is certified that the work contained in this thesis entitled “**AI Powered Voice-Guided Presentation: Revolutionizing PPT Explanations**” submitted by Mohit Ranjan (2012109), Rituraj Chanda (2012157) and Shibam Debnath (2012181) for the B.Tech. End Semester Project Examination, May, 2024 is absolutely based on his own work carried out under my supervision.

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“Learn everything that is good from others, but bring it in, and in your own way absorb it; do not become others”

- Swami Vivekananda

Abstract

The advent of Artificial Intelligence (AI) has precipitated a paradigm shift across various sectors, ushering in a new era of innovation and efficiency. Within this transformative landscape, our project emerges as a pioneering endeavor to leverage the full potential of AI to elevate the educational experience. At its core, our initiative introduces a groundbreaking AI-powered PowerPoint (PPT) presenter, designed not only to adeptly parse the content of presentations but also to dynamically generate a script for an AI model to articulate. This synthesis of advanced technologies, including text summarization and script generation, seeks to overcome the limitations inherent in traditional PPT presentations, particularly in the context of providing tailored and personalized learning experiences. By incorporating text summarization techniques, our AI presenter distills complex information into concise, comprehensible segments, ensuring that learners receive a nuanced understanding of the subject matter. The script generation component further enhances the adaptability of our system, enabling it to cater to diverse learning styles and preferences. This integration empowers our AI model not only to generate scripts but also to articulate them in a natural and engaging manner. The significance of this feature becomes apparent when considering its potential to make educational content accessible to visually impaired learners. In doing so, we aim to break down barriers and foster inclusivity in education, ensuring that all students, regardless of physical abilities, can partake in the learning process. Motivating our project is a profound awareness of the inadequacies inherent in current educational models, particularly in meeting the diverse needs of learners, especially those in economically challenged and remote areas. Recognizing the disparities in access and affordability, our vision is to create an AI-based teacher available around the clock. Through the seamless integration of AI technologies, our project envisions a future where education is not only democratized but also personalized, ensuring that every learner can unlock their full potential.

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Contents

Declaration	
Certificate	
Abstract	
Acknowledgements	
List of Figures	
List of Tables	
1. Introduction	01
2. Literature Survey	02
3. Motivation	07
4. Proposed Methodology	09
4.1 Content generation	09
4.1.1 Text extraction model	09
4.1.2 Llama 2 with prompt engineering model	10
4.1.3 Gemini pro with prompt engineering model	12
4.1.4 Fine tuned Llama 2 model	13
4.2 Text to speech	15
4.3 Lip-Sync Model Video Generation	16
4.4 Web Integration	18
5. Evaluation and Result	13
5.1 Subjective Rating Evaluation	22
5.2 Comparison of content generation models	23
5.2.1 Basic text extraction model	23
5.2.2 Llama 2 with prompt engineering model	23

5.2.3 Gemini pro with prompt engineering model.	23
5.2.4 Fine tuned Llama 2 model.	24
5.3 Google Text-to-Speech Integration	29
5.4 Lip-Sync Model Video Generation model	29
6. Web Application and Interface.	32
6.1 User-login system.	34
6.2 User interface.	34
6.3 Database integration.	35
6.4 Local network deployment..	35
7. Conclusion and Future Work	36
8. References	37
 Appendix I	 39
Appendix II	41

List of Figures

1. LLAMA-2-fine tuned chat model	02
2. Text Generation Flow	04
3. Text to Speech model	05
4. Transformer architecture and LLama architecture	10
5. Gemini Pro multi model architecture	12
6. Parameter efficient fine tune llama 2 model	14
7. Wav2Lip model Architecture	16
8. Client Server architecture	18
9. Flask as backend architecture	19
10. User satisfaction vs model name for easy PPTS	26
11. User satisfaction vs model name for medium PPTS	27
12. User satisfaction vs model name for Hard PPTS	27
13. Overall user satisfaction Vs model name	28
14. Comparison of all the models on three categories of PPTs	28
15. Featuring Questions Vs Manual Evaluation Rating	31
16. Frontend and Backend codebase	33

List of Tables

1. Manual Evaluation data of Text Generation Models	25
2. Mean Opinion Score for Text Generation Models	26
3. Manual Evaluation data of Lip-Sync models	30
4. Tech Stack of Web Integration	32

CHAPTER 1

Introduction

In today's education world, everyone uses PowerPoint presentations to share information. But the usual way of doing it doesn't always meet the different needs of students. Our project wants to change this by using AI to make educational content clearer and more accessible. We start by looking at why regular PowerPoint presentations may not be good enough, talking about the problems they have and why we need learning that is more personal and doesn't cost a lot. We explain our plan for fixing these issues in the introduction, where we talk about our main goals and how we're going to tackle the problems we found.

Regular PowerPoint presentations are everywhere in education, but they don't always work well for everyone. Some students learn better in different ways, and the usual presentations don't take that into account. Our project wants to use AI to make educational content better for everyone. Let's look at why regular PowerPoint presentations have problems. First, they don't always fit the way students learn. Some students need more personalized learning, something that speaks to them directly. Second, the usual way of doing things can cost a lot. This is a big issue, especially for students who may not have a lot of money.

Our project aims to revolutionize educational content presentation through AI integration. Traditional methods like PowerPoint presentations often fail to accommodate diverse learning needs, hindering effective learning. Our goal is to leverage AI to personalize learning experiences, making them accessible and affordable for all students. The paper proceeds with a Literature Survey (Section II), followed by Motivation (Section III), followed by the Proposed Methodology (Section IV). Performance evaluation is discussed in Section V, while experiment results and discussions are presented in Section VI. The paper concludes with insights and future directions in Section VII, and references are provided in Section VIII.

CHAPTER 2

Literature survey

The integration of Artificial Intelligence (AI) into educational technologies has witnessed substantial growth in recent years, with a focus on enhancing learning experiences and addressing the diverse needs of students. In this literature survey, we explore the key areas relevant to our project, including AI in education, text summarization, script generation, and Text-to-Speech (TTS) technology.

2.1 Llama 2: Open Foundation and Fine-Tuned Chat Models:

The paper [1] introduces the development and release of Llama 2, a collection of pretrained and fine-tuned large language models tailored for dialogue applications. Llama 2 is demonstrated to outperform existing open-source chat models, showcasing superior performance metrics and response coherence. LLAMA-2 fine tuned chat model shown in fig 1. The paper [1] delves into the fine-tuning methodology, emphasizing safety improvements to mitigate biases and enhance model robustness. Llama 2's open foundation is highlighted, inviting community collaboration and contributions. The authors stress the importance of responsible AI development and provide detailed insights into the technical aspects, including model architecture, training data, and methodologies, offering a valuable resource for researchers and practitioners in the field of natural language processing, as shown in the Fig 1.



Fig.1 : LLAMA-2-fine tuned chat model

2.2 Gemini Pro

Gemini [6], Google's advanced large language model (LLM), stands on the shoulders of giants in the field. Here, we explore key research areas that inform its capabilities.

1. Multimodality: A core strength of Gemini is its ability to understand and reason across different data types (text, code, images). This aligns with research on multimodal learning in LLMs. Examining research in this area likely influenced Gemini's development, allowing it to integrate information from various sources for richer reasoning.

2. Responsible AI: Gemini's design prioritizes safe and ethical use, reflecting the growing emphasis on responsible AI development. Research delves into this crucial area, exploring best practices and potential risks. Examining such research likely informed how Google safeguards against potential risks associated with powerful AI like Gemini.

3. Fine-tuning Techniques: Gemini utilizes supervised learning and reinforcement learning with human feedback to refine its outputs. This aligns with advancements in fine-tuning LLMs. Studying research in this area likely contributed to Gemini's learning process, shaping it to prioritize helpfulness and safety in its responses.

By examining these and similar research areas, we gain valuable insights into the foundations of Gemini. This exploration highlights the crucial role of ongoing research in shaping powerful, versatile, and responsible AI tools like Gemini.

2.3 Text Generation:

Text generation is a dynamic field witnessing substantial contributions that leverage innovative approaches to enhance the quality and interactivity of generated content. Text generation flow diagram shown in fig 2. One notable paper, "*Interactive Text Generation*," [8] pioneers the introduction of an Interactive Text Generation task. This task allows training generation models interactively through user simulators, facilitating model guidance without direct user involvement. Employing Imitation Learning, the paper [8] demonstrates the superiority of interactively trained models over non-interactive counterparts, even when subjected to the same user input or edit budgets. This contribution enhances the understanding and applicability of interactive approaches in the realm of text generation.

In tandem, "*A Systematic Literature Review on Text Generation Using Deep Neural Network Models*" [2] offers a comprehensive overview of text generation research employing deep neural network models. Conducting a systematic literature review from 2015 to 2021, the paper[6] identifies key aspects such as language translation, dataset balancing, next word prediction, and various other domains. By cataloging 90 primary studies, the review not only

presents existing research but also identifies research gaps, providing valuable insights for future exploration. The paper's [2] contribution extends to offering guidelines for practitioners and proposing future directions to steer advancements in text generation using deep neural network models. Additionally, the "*Long Text Generation Challenge*" [3] introduces a unique shared task focused on generating consistent, human-like long texts. Proposing an innovative statistical metric, GAPELMAPER, to measure the structuredness of generated text, the paper [3] presents a holistic evaluation protocol that incorporates human assessment. This challenge encourages researchers to explore diverse sampling approaches, prompting strategies, and architectures, thereby addressing the limitations associated with generating lengthy texts. Collectively, these contributions not only advance the understanding of interactive training methodologies but also provide a comprehensive overview of existing research trends and foster new avenues for overcoming challenges in long text generation as shown in fig 2.

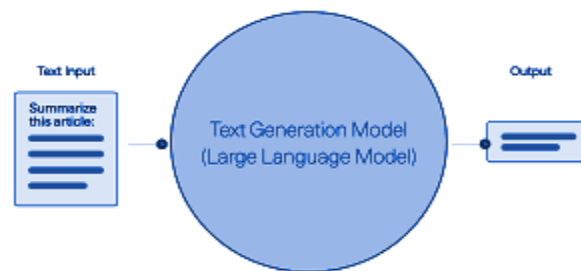


Fig. 2: Text Generation Flow

2.4 Text-to-Speech Technology:

There are now two types of TTS[15] conversion systems on the market, rule-based and machine learning-based. Rule-based TTS[16] systems generate voice from text based on a predetermined set of rules and algorithms. Text - to-speech model shown using fig 3. To generate speech that is convincingly human-like, these systems often consult a massive database of phonetic and prosodic information. High-quality speech is possible with rule-based TTS systems, but their development and upkeep can be time-consuming and costly due to the complexity of the requisite database. On the other hand, TTS systems based on machine learning employ statistical models to convert text into speech. The systems may be trained on massive datasets to produce high-quality speech, and they commonly use deep neural networks (DNNs) to represent the connections between text and speech. While they have their benefits, TTS systems that rely on machine learning are still susceptible to overfitting and generalization issues, and they are only as good as the training data they are given. Recent research has analyzed the current state of TTS conversion, and the limits of current TTS systems have been discussed [17]. The limitations of rule-based TTS systems, such as their need for a vast database of phonetic and prosodic information and its difficulty in modeling complex linguistic processes, were noted

by Liu et al. (2018) [11] in their review of TTS[18] systems. They also talked on the challenges of modeling long-term dependencies in speech and the need for high-quality training data machine learning-based TTS systems. In a similar vein, Tacchini et al. (2019) [12] reviewed deep learning-based TTS systems and noted the shortcomings of current TTS systems, such as the need for massive volumes of annotated speech data and the challenge of accurately modeling prosodic variation and expressiveness. They also talked about the issues associated with training deep neural networks for TTS conversion, such as the necessity for a lot of computational resources and the difficulties of avoiding overfitting and generalization problems. Text to speech model function is shown in fig 3.



Fig. 3: Text to Speech model

2.5 Lip-Sync Video Generation:

Lip-syncing, the technique of matching a person's lip movements to pre-recorded audio, has been around for decades. Traditionally, it involved meticulous timing and coordination between the performer and the audio playback. This technique is commonly used in films where sound recording might be difficult during filming, or in live performances where elaborate dance routines might make singing challenging.

However, recent advancements in deep learning have introduced more sophisticated approaches to lip-syncing. One such technique is Wav2Lip, presented in the paper "Generating Lip Synchronized Videos from Audio" by Prekumar et al. (2020). Wav2Lip tackles the challenge of creating realistic lip-syncing in everyday videos, offering significant advantages over traditional methods.

2.5.1 Wav2Lip: Deep Learning for Realistic Lip-Syncing

Wav2Lip [19] utilizes a deep learning architecture to animate lip movements in pre-recorded videos (of any person) to match a target audio clip as discussed in the paper[19]. This makes it applicable to a wide range of scenarios, from generating training data for chatbots to creating

special effects in movies. Unlike traditional methods that require precise timing, Wav2Lip can handle unconstrained videos, where the original video doesn't match the target audio.

The secret weapon behind Wav2Lip's success lies in its deep learning architecture. A key component is a specially trained "lip-sync discriminator" network as explained in the paper [19]. This network acts as a critical evaluator, assessing the quality of generated lip movements. By continuously evaluating its own outputs against this discriminator, Wav2Lip refines its ability to produce highly realistic lip-syncing that closely follows the spoken words. This innovative approach allows Wav2Lip to achieve impressive results, nearing the quality of real synced videos.

CHAPTER 3

Motivation

The project on AI Powered Voice-Guided Presentation: Revolutionizing PPT Explanations is based upon multiple goals. Starting from Helping everyone learn better, supporting teachers for better education, Efficiently and saving resources etc. The key motivations behind the project is explained below:-

3.1. Helping Everyone Learn Better:

We want to make a tool so that everyone, including those who can't see well, can learn easily. It's about making sure nobody feels left out when it comes to learning new things. Our project aims to make quality education accessible for all, even to the people staying in the remotest part of the country, who have very less resources and can't afford quality education. Sometimes presentations are hard for people who can't see well. Our project wants to make sure learning is easy for everyone, no matter what challenges they might have.

3.2. Supporting Teachers for better Education:

Teachers already have a lot on their plates. With our system, we want to take care of some routine tasks, allowing teachers to focus on creating engaging and personalized learning experiences for their students. Imagine a tool that ensures every student, no matter who the teacher is, receives a consistently high-quality learning experience.

3.3. Efficiency and Saving Resources:

For schools and organizations, resources are often limited. Our project aims to use resources efficiently, making sure that every rupee spent on education delivers the maximum impact. Technology should be about the future. Our project is not just a quick fix; it's a long-term approach to education that considers scalability and adaptability for the future.

3.4. Bringing Education to Remote Places:

Our project is about making sure that students in very remote areas have the same chance to learn and explore as students in big cities. It's about giving everyone, no matter where they are, an equal shot at a great education. Sometimes, students in faraway places feel left out because they don't have the same resources. Our project wants to change that. It wants to bring the excitement of learning to every corner, so no student feels left behind.

In conclusion, a research study on AI Powered Voice-Guided Presentation Revolutionizing PPT Explanations is motivated by a wide range of real-world applications, from Education and assistive technology to consumer electronics and urban planning. In addition we use artificial intelligence, for optimum use of resources and provide better education facilities in rural areas

CHAPTER 4

Methodology

The goal of this research is to revolutionize PowerPoint presentations by incorporating AI powered voice guidance. The methodology involves a multi-step process, combining natural language processing, text extraction, natural language understanding, text generation, speech synthesis to enhance the explanation of each slide's content and then finally lip-Sync-Video generation to make it more user friendly, visuals create more impact on learning. The complete methodology we followed to progress in this project can be divided into four main segments, which are text-extraction, text-generation,, text-to-speech and Lip-Sync Model Video Generation and finally web integration of all these. The detailed explanation of each process is given below:

4.1. Content Generation

In this part we generate the content said by the avatar to explain the given PPT to the user. The main components under this section are text-extraction, text-generation, content-segmentation and text-summarisation. The detailed explanation of each component is given below

4.1.1. Text Extraction:

The process of extracting contents from a PowerPoint (PPT) file involves using the `'python-pptx'` library in Python. The script begins by creating a `'Presentation'` object to represent the PPT file. It then iterates through each slide in the presentation. For each slide, it determines the heading by checking if the slide has a title; if not, it defaults to a generic heading format. The script then traverses each sh within the slide, accumulating text content from shapes that have text attributes. This text content, along with the heading, is organized into a dictionary, where the heading serves as the key. Finally, the script returns this dictionary containing the extracted slide contents. The approach is modular, allowing for flexibility in handling different PPT structures, and the use of conditional statements ensures robust extraction even when specific elements, like titles, may be absent.If in case of pdf we can use pip library and langchain.

4.1.2. Text Generation Using Llama2-7B-Chat-Model:

The LLaMA-2-7B Chat Model is a language model with an impressive 7 billion parameters. It is based on the transformer architecture, which excels at capturing contextual information and handling long-range dependencies.

4.1.2.1. Architecture Components:

The LLaMA-2-7B Chat Model is built upon a **transformer-based architecture**, which is a type of neural network architecture that has revolutionized the field of natural language processing (NLP). The LLaMA-2-7B Chat Model's transformer-based architecture is designed to handle sequential data, making it ideal for a wide range of NLP tasks. At its core, the transformer model relies on a self-attention mechanism that allows it to process each token in the context of all other tokens in the sequence. This is a departure from earlier sequence modeling approaches like RNNs and LSTMs, which processed tokens in a linear fashion. Transformer model Architecture shown in fig 4.

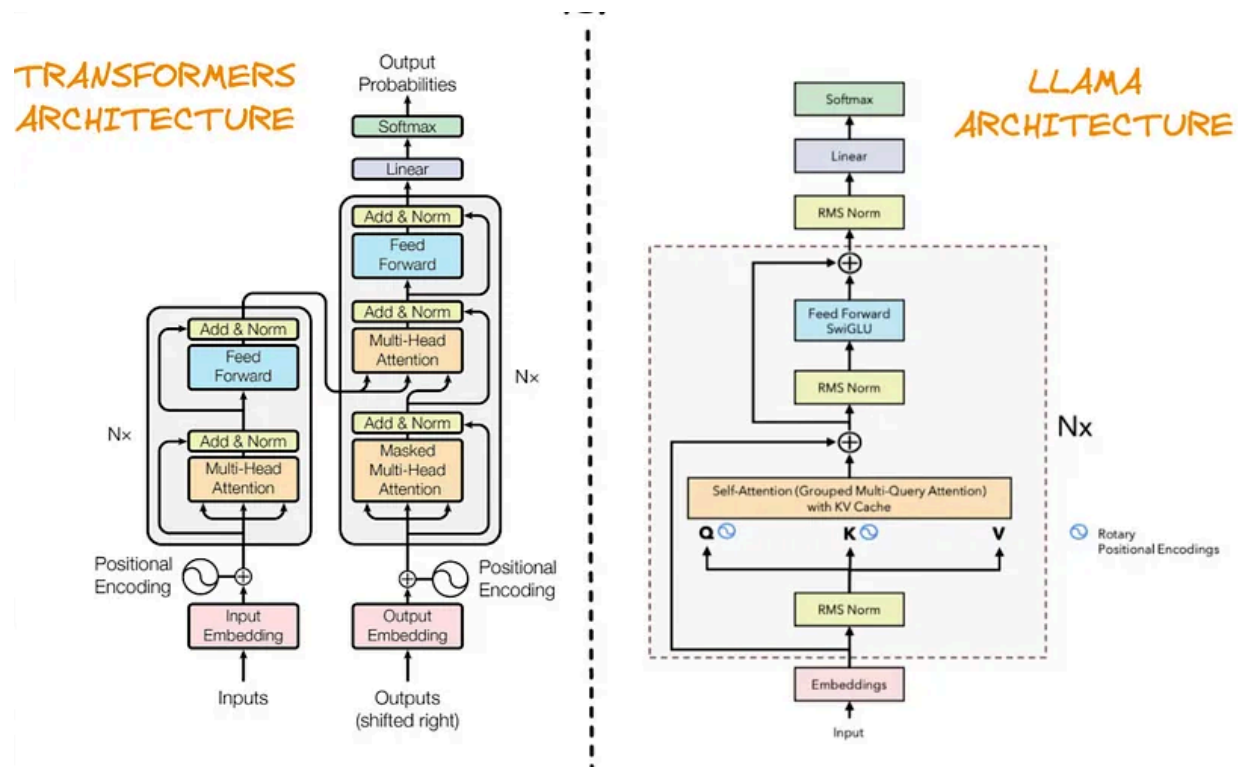


Fig. 4 : Transformer Architecture vs Llama Architecture

In a transformer, the input sequence is first converted into a set of embeddings, which are then passed through several layers of the transformer network. Each layer consists of two main sub-layers: the multi-head self-attention mechanism and a position-wise feed-forward network.

Multi-head attention is a key feature of the transformer architecture, allowing the model to focus on different positions of the input sequence simultaneously. This enables the model to capture a richer understanding of the context. The attention heads work in parallel, each potentially focusing on different aspects of the sequence, and their outputs are combined and linearly transformed into the expected dimension. **Positional encodings** are added to the input embeddings to give the model information about the order of the tokens. Unlike RNNs, which inherently consider the sequence order, transformers need this additional information to maintain the sequence context. The **feed-forward neural networks (FFNs)** within each layer are fully connected layers that apply transformations to the intermediate representations. These networks are identical across different positions but use different parameters from layer to layer.

Layer normalization is applied before each sub-layer (self-attention and FFNs) and residual connections are used after each sub-layer, which helps in stabilizing the gradients during training and allows for deeper models by preventing the vanishing gradient problem. Finally, the output layer of the transformer uses a softmax function to generate a probability distribution over the vocabulary for the next token in the sequence. This output can then be used for various tasks such as text generation, translation, summarization, and more.

The LLaMA-2-7B's transformer-based architecture, with its self-attention and multi-head attention mechanisms, positional encodings, layer stacking, FFNs, and normalization techniques, provides a powerful framework for generating coherent and contextually relevant text, making it highly effective for a multitude of natural language processing applications.

4.1.2.2. Prompt Engineering Strategies:

Prompt engineering is a crucial aspect of leveraging the LLaMA-2-7B Chat Model effectively across various natural language processing tasks. It involves crafting inputs or prompts that guide the model towards producing desired outputs. Contextual prompts offer relevant context to the model, helping it understand the task at hand. For instance, providing opening sentences from an article can guide the model in generating a summary. Task-specific prompts tailor the input to the specific task being performed. For example, using "Explain the following text:" followed by the input directs the model to Explain the given text.

Temperature and top-k sampling parameters are key in controlling the diversity and determinism of the generated text. Adjusting these parameters during text generation allows for fine-tuning the output according to preferences. Prefixes or constraints are also valuable in guiding the initial part of the generated text. For instance, using "In summary, the key points are:" as a prefix can prompt the model to list important points. By continually evaluating its output and adjusting strategies as needed, users can ensure that the model stays effective and aligned with their requirements. Overall, prompt engineering strategies play a vital role in optimizing the performance of the LLaMA-2-7B Chat Model across a wide range of tasks.

4.1.2.3. Evaluation and Iteration:

Experiment with prompts, fine-tuning strategies, and hyperparameters. Continuously refine your approach based on results. LLaMA-2-7B's flexibility allows you to tailor it to specific use cases. Explore different prompt variations and observe how they impact output quality.

4.1.3. Text Generation with Gemini-pro:

The Gemini-Pro LLM model, a cutting-edge development in language models, is built on a transformer-based architecture that enables it to handle complex natural language processing tasks. the architecture is shown in fig 5.

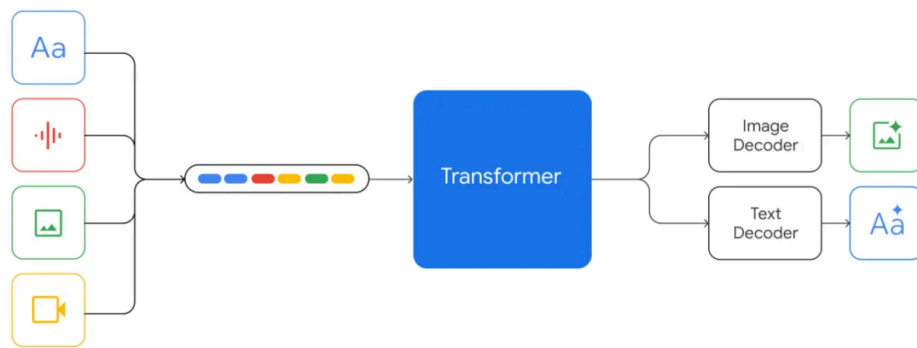


Fig.5 : Gemini pro multimodal architecture

Transformer-Based Architecture: At the heart of Gemini-Pro is the transformer architecture, which is renowned for its ability to process sequences of data, particularly text[6]. This architecture is composed of an encoder and decoder, each consisting of multiple layers that transform the input data into a more useful representation.

Encoder: The encoder's primary function is to process the input data and create a context-rich representation. It does this through several layers, each containing two sub-layers:

- A self-attention mechanism that allows the model to weigh the importance of different parts of the input data.
- A feed-forward neural network that transforms the self-attended features into a higher-level representation.

Decoder: The decoder uses the encoder's output to generate the final text output. It also has multiple layers, each with:

- A masked self-attention mechanism that prevents the model from seeing future tokens in the sequence, ensuring that the generation process is autoregressive.

- An encoder-decoder attention mechanism that allows the decoder to focus on relevant parts of the input sequence.
- A feed-forward neural network similar to the one in the encoder.

In summary, the Gemini-Pro LLM model's transformer-based architecture as shown in figure 4, combined with its multimodal capabilities and advanced prompt engineering techniques, makes it a powerful tool for generating coherent and contextually relevant text while explaining the contents of our PPT. Its design reflects the latest advancements in AI and NLP, offering a glimpse into the future of language models.

The process of extracting text from a Gimli file likely involves utilizing a custom parser or library designed to handle Gimli's specific structure. Here's a general outline:

Gimli File Parsing The initial step involves loading the Gimli file using appropriate functions or libraries. This might involve creating a reader object specific to Gimli files. **Content Segmentation** Gimli files might have a unique structure for storing text content. The parser would need to identify and segment the text data from other elements within the file. This could involve recognizing delimiters, markers, or patterns specific to Gimli's text storage format.

Text Extraction and Organization Once the text segments are identified, the parser would extract the actual text content. For instance, it could be stored in a list where each element represents the text from a specific section or element within the Gimli file. Alternatively, a dictionary structure could be used where keys represent sections or elements, and values hold the corresponding extracted text.

Additional Considerations **Leveraging Existing Libraries:** If there are existing libraries or tools designed for Gimli file manipulation, the text extraction process could be simplified by utilizing their functionalities. These libraries might provide pre-built functions for parsing Gimli files and extracting specific data elements, including text content. **Customization:** The specific methodology for text extraction might need to be customized based on the exact structure and format of Gimli files. Different Gimli variations might require adjustments in the parsing and segmentation steps to handle their unique characteristics.

4.1.4. Text Generation with LLaMA-2 after fine tuning:

To begin text generation with a fine-tuned LLaMA-2-7B model using a custom dataset, the first step is data collection and preprocessing. This involves gathering domain-specific data relevant to the target task, such as articles, documents, or other textual sources. Subsequently, the collected data is preprocessed to ensure consistency and quality. This includes tasks such as tokenization, lowercasing, and removal of irrelevant information, ensuring that the data is in a suitable format for training the model.

Fine-tuning the LLaMA-2-7B model for text generation is a comprehensive process that significantly enhances the model's proficiency in generating text specific to various tasks or domains. The methodology commences with the meticulous preparation of the dataset, which is divided into training, validation, and test sets. These sets are meticulously encoded to align with the model's tokenization and formatting specifications. The fine-tuning journey begins with the model's initialization using its pre-trained weights, leveraging the foundational language understanding it has already acquired.

As the fine-tuning progresses, the model undergoes parameter updates on the prepared dataset, aligning its capabilities more closely with the targeted domain or task. This phase employs innovative techniques such as Layer-wise Only Relevant Attention (LoRA)[16] and Quantized LoRA (QLoRA). LoRA refines the attention mechanism of the transformer layers for greater parameter efficiency by introducing trainable low-rank matrices that adapt the self-attention's weight matrices. QLoRA extends this by incorporating quantization[17], reducing the precision of the weights to achieve faster computation and a more compact model size, with minimal impact on performance. Hyperparameter tuning is another critical aspect, where parameters such as learning rate, batch size, and training epochs are adjusted.

The approach of Parameter Efficient Transfer Learning is particularly noteworthy, focusing on updating a select subset of the model's parameters to streamline the fine-tuning process.[18] This is especially beneficial when working with limited computational resources or smaller fine-tuning datasets. Parameter Efficient Fine tuning steps are shown in fig 6.

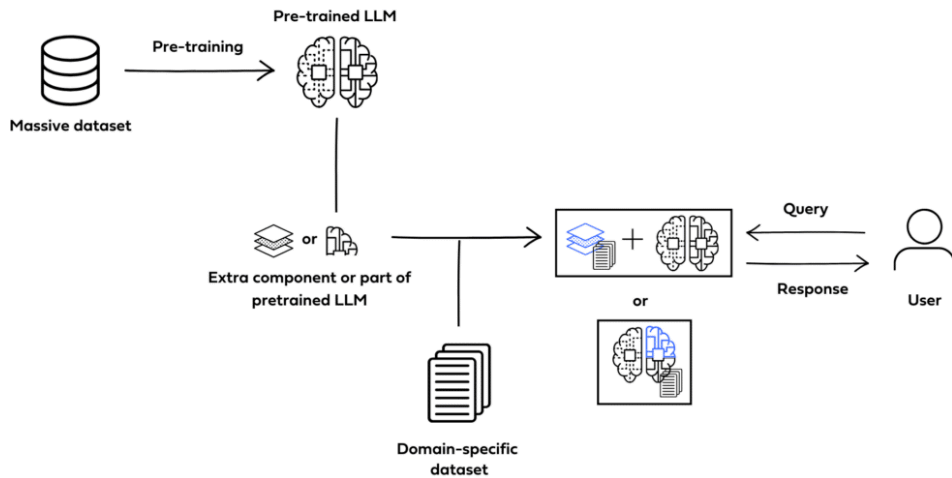


Fig. 6: Parameter Efficient Fine-Tuning of LLama 2 model.

For implementation, a suite of tools and libraries, including Hugging Face's Transformers and Parameter Efficient Fine-tuning Techniques (PEFT), provide the necessary infrastructure to apply LoRA, QLoRA, and other fine-tuning methods effectively. By adhering to this detailed methodology, the LLaMA-2-7B model is fine-tuned to produce text outputs that are not only highly relevant to the user's prompts but also demonstrate enhanced performance on specific

natural language tasks. The strategic use of LoRA, QLoRA, and Parameter Efficient Transfer Learning ensures that the fine-tuning is both efficacious and efficient, culminating in a model that is exceptionally well-suited for its intended applications.

For effective text generation, appropriate prompt engineering strategies are chosen. These may include contextual prompts, task-specific prompts, temperature, and top-k sampling, all of which guide the model's output towards the desired results. Evaluation of the fine-tuned model's performance is conducted using appropriate metrics for the text generation task, such as ROUGE for summarization or BLEU for translation. The generated text is compared against human-written references to assess quality and coherence.

Following evaluation, an iterative refinement process is undertaken. This involves adjusting hyperparameters and prompt engineering strategies based on evaluation results, fine-tuning the model further with additional data or iterations to improve performance. Once the model is deemed ready, it is deployed into production for text generation tasks. Continuous monitoring of the model's performance in real-world scenarios is crucial, with feedback gathered from users for further refinement. The model is then updated and adapted based on this feedback and changing requirements to ensure ongoing effectiveness.

4.2. Text to speech

In our ongoing efforts to make information accessible to everyone, we've introduced Google Text-to-Speech (GTTS)[19] to our platform. This tool helps convert written text into spoken words, making content more accessible, especially for those who prefer listening or have visual impairments. Within the domain of Text-to-Speech (TTS) technology, the incorporation of the Google Text-to-Speech API is a crucial element in our project, delivering an unmatched auditory experience. Harnessing the capabilities of Google's advanced AI technologies, this API provides a comprehensive suite of features that enhance the quality of synthesized speech to a level approaching human-like precision. The API[20] is distinguished by its high-fidelity speech, a key attribute influenced by DeepMind's proficiency in speech synthesis. This characteristic imparts a natural intonation to the voices, thereby enriching the user experience and rendering educational content more captivating.

Why GTTS Matters: GTTS has a few important features that make it valuable for our mission:

Natural Sounding Speech : GTTS doesn't just read text aloud; it makes it sound natural, almost like a person speaking. This makes the information more engaging and easier to follow.

Language Variety: It supports many languages and accents, so our users can experience content in the language they're most comfortable with. This inclusivity is crucial for our diverse community.

How We Use GTTS: Our platform seamlessly integrates GTTS to convert written content into spoken words. Whether users are on a computer, tablet, or phone, they can access information more conveniently through voice.

4.3. Lip-Sync Model Video Generation:

We employed a wav2lip model for lip-sync video generation which uses CNN and GAN for creating realistic videos[15]. Wav2Lip stands out in the world of deep learning as a model dedicated to creating realistic lip movements in sync with any audio source. Imagine having the ability to take a silent video and make it appear as if the person is speaking new words – that's the magic of Wav2Lip.

Wav2Lip achieves this impressive feat through a clever analysis process. It dissects the audio, typically converting it into a mel-spectrogram, which acts like a fingerprint of the speech patterns. This allows the model to understand the nuances of the audio, from the rise and fall of pitch to the distinct sounds produced by different pronunciations. With this knowledge in hand, Wav2Lip meticulously generates corresponding lip movements in the video frame by frame.

However, Wav2Lip takes a unique path compared to some models. It operates directly on the two-dimensional image data (2D-based approach) of the video frames. This makes it computationally efficient, allowing for faster processing. While this approach prioritizes speed, it can introduce subtle limitations in visual quality compared to 3D methods. Architecture of Wav2Lip is shown in fig 7.

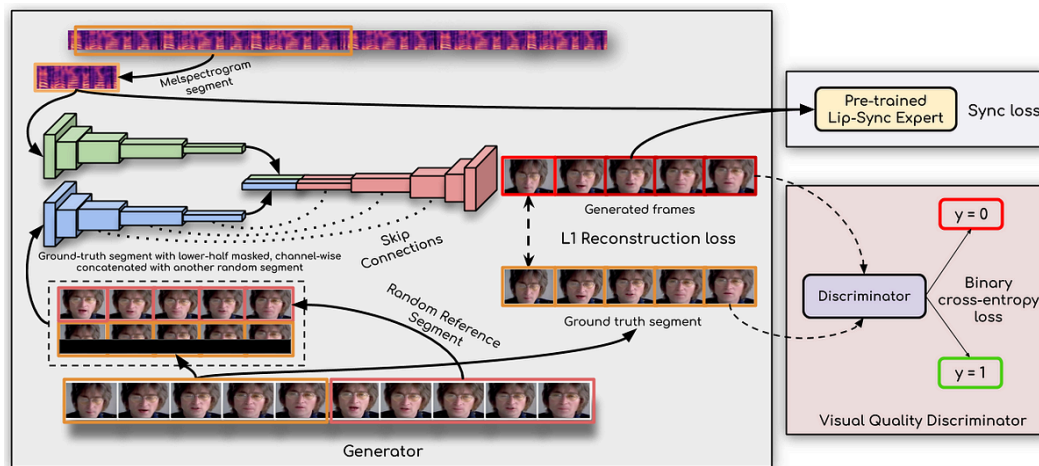


Fig. 7 : Wav2Lip Model Architecture

Let's delve deeper into Wav2Lip's internal workings. The process can be broken down into three key stages:

4.3.1 Encoding the Information:

Imagine Wav2Lip as a master decoder, but first it needs to understand the languages it's working with. Here's where the encoding comes in. A video encoder takes the current frame from the video (with the lower face conveniently masked out to focus on the non-moving parts) and a randomly chosen reference frame from the same video. It analyzes these frames and compresses their key features into a latent representation – a sort of hidden code capturing the essence of the visual information. On the audio side, another encoder takes the mel-spectrogram, the fingerprint of the speech, and encodes it into its own latent representation.

4.3.2 The Decoding Dance:

Now that Wav2Lip has a grasp of both the visual and audio information in their latent forms, it's time for the magic to happen. A decoder, the mastermind behind the lip-syncing, takes the stage. It receives the latent representations from both the video and audio encoders. Acting like a skilled translator, the decoder combines this information. It understands the visual context from the video and the speech patterns from the audio. With this combined knowledge, it meticulously generates a brand new video frame where the lip movements perfectly align with the audio.

4.3.3 Training for Perfection:

Just like any skilled performer, Wav2Lip gets better with practice. It's trained on a massive dataset of videos where the audio and visuals are already perfectly synchronized. During training, the model constantly refines its ability to map the features extracted from the audio (remember the mel-spectrogram?) to lip movements that flawlessly match the ground truth – the original video with its natural lip movements. Wav2Lip doesn't operate in a vacuum. To ensure the generated lip movements are realistic and the overall video frame maintains its natural look, the training process incorporates a quality check. It uses a "lip-sync expert" – a component trained to identify natural-looking lip movements – and a "visual quality discriminator" – a component that assesses the overall visual quality of the generated frame. These checks guide Wav2Lip towards generating the most convincing results possible.

Wav2Lip offers a choice between two pre-trained models, each catering to a specific need:

Wav2Lip: This model prioritizes achieving the most accurate lip-sync possible, ensuring the generated movements precisely match the audio.

Wav2Lip + GAN: This model incorporates a Generative Adversarial Network (GAN) during training. A GAN is like a competitive environment where two models work against each other. In this case, one model tries to generate realistic video frames, while the other tries to distinguish the generated frames from real ones. This competitive training process helps Wav2Lip + GAN generate video frames with superior visual quality, though it may lead to slightly less precise lip-sync compared to the standard Wav2Lip model.

4.4 : Web Integration:

In the realm of educational technology, the successful integration of web applications can significantly enhance the accessibility and user experience of AI-powered tools. Our project seamlessly marries Flask, a Python web framework, and React, a popular JavaScript library for building user interfaces, to create a robust and user-friendly web application. This section delves into the intricacies of our web integration, outlining the architecture, functionality, and the seamless interaction between the Flask server and React components.

4.4.1 Architecture Overview:

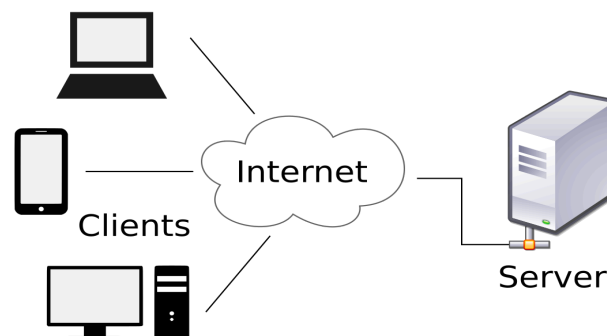


Fig. 8: client server architecture

The web application architecture is designed with a client-server model, with Flask serving as the backend server and React handling the frontend interactions. This approach ensures a clean separation of concerns, allowing for scalability, maintainability, and efficient data flow. The Flask server provides RESTful API endpoints, enabling communication between the backend and frontend components. Client-server architecture has become a cornerstone of modern computing due to its ability to efficiently distribute tasks and resources across a network. Here's a breakdown of its enduring fame: The Client Server architecture shown in above fig 8.

Division of Labor: Client-server splits functionalities between clients (user devices) and servers (powerful machines). Clients handle user interaction and basic processing, while servers

manage data storage, complex calculations, and security. This division optimizes resource utilization – clients don't need powerful hardware, and servers can efficiently serve multiple clients.

Scalability: Adding new clients is simple – the server's capacity can be increased to handle the additional load. This makes client-server ideal for applications with growing user bases, like web services and enterprise systems.

Centralized Management: Data and applications reside on the server, enabling centralized administration and security updates. IT teams can easily maintain and update software on the server, ensuring consistency across all client devices.

Improved Security: Since sensitive data resides on the server, security measures can be concentrated there. This provides a more robust defense against unauthorized access compared to individual client-side security.

4.4.2 Flask Server:

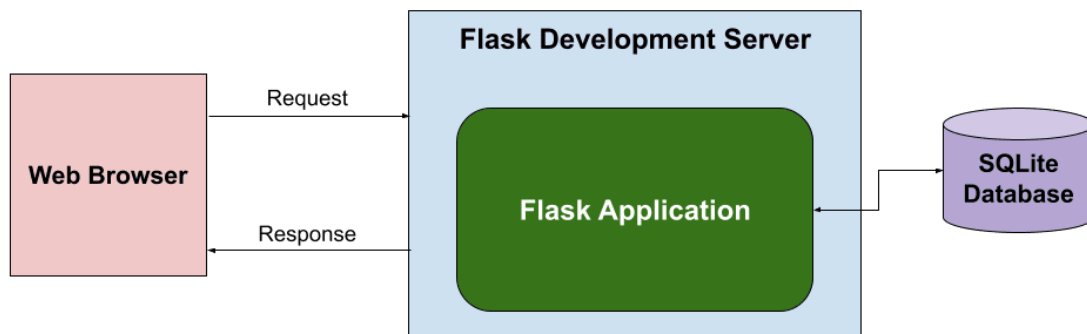


Fig. 9: Flask as backend architecture

Our Flask server acts as the backbone of the web application, handling key functionalities such as authentication, data processing, and communication with external APIs. Flask backend architecture shown in fig 9. Different endpoints are meticulously crafted to cater to specific functionalities, ensuring a modular and organized codebase. For instance, authentication endpoints manage user login and registration, while others handle PPT file uploads, text summarization, script generation, and Text-to-Speech functionalities. The Flask server acts as the bridge between the user interface and the AI-powered backend. Flask is a popular Python web framework renowned for its simplicity and flexibility. Let's delve into its core strengths and explore why it's a favorite among developers.

Microframework Magic: Unlike bulky frameworks, Flask embraces a minimalist approach. It provides a core set of features, allowing developers to choose additional functionalities through extensions. This streamlined approach fosters rapid development and keeps codebases clean.

Routing the Way: Flask excels at routing – the process of directing incoming web requests to appropriate functions within your application. Using decorators, you can easily define URL patterns and link them to specific Python functions that handle those requests.

Templating Made Easy: Flask integrates seamlessly with Jinja2, a powerful templating engine. Jinja2 allows you to create dynamic web pages with HTML-like syntax and embedded Python code. This separation of concerns improves code readability and maintainability.

Development Friendly: Flask prioritizes developer experience. Its built-in debugger simplifies troubleshooting, and its comprehensive documentation provides clear guidance. This makes Flask ideal for beginners looking to enter the world of web development.

Real-World Applications: Flask's versatility shines in diverse projects. It's used for crafting APIs, microservices, data dashboards, lightweight web applications, and even prototyping complex web architectures.

4.4.3 React Frontend:

The React frontend complements the Flask backend, providing an interactive and intuitive user interface. Leveraging React's component-based architecture, we design modular components for seamless navigation and an engaging learning experience. The user interface incorporates elements for uploading PPT files, configuring user settings, and interacting with the AI-generated educational content. React's state management ensures dynamic updates, providing real-time feedback to users.

Component-Based Architecture: React breaks down UIs into reusable components, each managing its own state and logic. This promotes modularity, simplifies code organization, and facilitates maintenance. Changes made to a component only affect its specific section, keeping the rest of the UI intact.

Virtual DOM for Efficiency: React utilizes a virtual DOM, a lightweight representation of the actual DOM (Document Object Model) that manages your web page's structure. When changes occur, React compares the virtual DOM with the actual DOM and updates only the necessary parts, resulting in efficient and smooth UI updates.

JSX for Readability: JSX (JavaScript XML) extends JavaScript syntax, allowing you to write HTML-like structures directly within your React code. This improves readability and makes it easier to visualize how your UI components will translate to the final web page.

Unidirectional Data Flow: React promotes a unidirectional data flow, where data is passed down from parent to child components through props (properties). This simplifies reasoning about how data affects different parts of the UI and helps prevent unintended side effects.

State Management for Complexity: For applications with complex data management needs, React offers solutions like Redux or Context API. These tools facilitate managing application state across multiple components and maintaining consistency.

Virtual Community and Ecosystem: React boasts a vast and active community of developers. This translates to an extensive ecosystem of libraries and tools, empowering developers to tackle various challenges and build feature-rich web applications with relative ease.

In conclusion, the web integration of our AI-powered educational platform using Flask and React represents a harmonious synergy between backend functionality and frontend user experience. This architecture enables a dynamic, real-time, and secure learning environment, fulfilling the project's objectives of accessibility, personalization, and user engagement.

CHAPTER 5

Evaluation and Results

Presenting the tangible outcomes of our project, this section discusses the results obtained from testing and user feedback. We analyze the impact on personalized learning experiences, accessibility for visually impaired learners, and the overall effectiveness of our AI-powered educational platform. The discussion delves into the implications of the results and potential areas for improvement. We have divided our result and analysis part into three different sections namely

1. Content Generation
2. Text to Speech
3. Lip-Sync Video Generation

5.1 Subjective Rating Evaluation

At the heart of our evaluation methodology lies the fundamental principle of subjective rating evaluation, an indispensable approach that capitalizes on human perceptual faculties to meticulously gauge the quality of our models. Through the solicitation of feedback from a carefully curated cohort of 20 individuals, each meticulously tasked with the critical endeavor of immersing themselves in video outputs produced by distinct models, we meticulously cultivated a rich and diverse corpus of subjective assessments. Employing a nuanced scale ranging from 1 to 10, participants were entrusted with the solemn duty of scrutinizing the fidelity, naturalness, and intelligibility encapsulated within the synthesized voices. In this immersive auditory journey, participants navigated the intricate nuances of tone, cadence, and articulation, each note resonating with the essence of user experience. As custodians of user perception and satisfaction, our esteemed participants bestowed upon us invaluable insights, illuminating the subtle contours

of model performance and unveiling the tapestry of human-machine interaction. Through this synergistic convergence of human judgment and computational prowess, we transcend the realm of mere metrics and metrics, embarking on a transformative odyssey towards the apotheosis of AI-driven educational excellence.

The scores provided by the evaluators were meticulously recorded and aggregated, yielding a comprehensive dataset encapsulating the subjective assessments of each model's output. Leveraging this dataset, an expected accuracy metric was computed, offering a quantitative measure of the models' performance in voice generation.

5.2. Comparison of content generation models :

In the pursuit of creating an AI-powered PowerPoint presenter acting as an AI teacher, our project underwent rigorous evaluation to ascertain the efficacy of various components, particularly focusing on content generation text-to-speech conversion and Lip-Sync-Video generation. The evaluation protocol was meticulously crafted to ensure robustness and reliability in assessing the performance of the implemented models.

5.2.1 Text Generation Models

The evaluation of the text generation models showcases a progressive enhancement in the sophistication and capabilities of each model. Here's a more detailed exploration of their roles in the content creation process:

5.2.2 Basic Text Extraction Model: This model serves as the groundwork for content extraction, utilizing elementary text extraction methods to capture information from PowerPoint presentations. Its primary function is to identify and retrieve textual content, which can include bullet points, captions, and paragraphs. The model's simplicity makes it fast and efficient, although it may lack the depth of understanding necessary for more complex explanations.

5.2.3 Llama2 with Prompt Engineering: Building upon the foundational model, Llama2 introduces advanced prompt engineering to the mix. This technique involves crafting prompts that guide the model to generate content that is not only contextually aware but also maintains coherence throughout the script. Prompt engineering enables Llama2 to produce summaries that are more aligned with the user's intent and the underlying narrative of the presentations. This model represents a significant step forward in generating summaries that are both informative and readable.

5.2.4 Gemini Pro with Prompt Engineering: At the apex of text summarization technology, Gemini Pro incorporates even more refined prompt engineering methodologies. This model is designed to understand the subtleties and complexities of the content, allowing it to generate scripts that are not just coherent and contextually relevant, but also nuanced and comprehensive.

Gemini Pro's ability to synthesize content at this level means it can produce scripts that are almost indistinguishable from those written by human experts, making it an invaluable tool for creating high-quality summaries.

Each model plays a vital role in the content creation pipeline, with the sophistication and capabilities increasing from the Basic Text Extraction Model to Gemini Pro. The use of prompt engineering is a game-changer in this process, enabling the models to produce outputs that are progressively more refined and tailored to the specific needs of content summarization. As the technology continues to evolve, we can expect even more advanced models that can handle a wider array of content creation tasks with greater precision and subtlety.

5.2.5 Fine-tuned LLaMA-2-7B Model: This model represents a tailored evolution of the LLaMA-2-7B, where it has been fine-tuned on a specific dataset to optimize its performance for particular tasks or domains. The fine-tuning process involves techniques such as LoRA (Layer-wise Only Relevant Attention) and QLoRA (Quantized LoRA), which are designed to make the model more parameter-efficient and adapt its attention mechanism to focus on the most relevant aspects of the input data. Additionally, Parameter Efficient Transfer Learning techniques are employed to update a small subset of the model's parameters, making the fine-tuning process more efficient and effective. The fine-tuned LLaMA-2-7B model excels in generating text that is not only coherent and contextually aware but also finely attuned to the nuances of the specific domain it has been trained on. This results in a model that can produce content with a higher degree of specificity and accuracy, making it an invaluable asset in scenarios where domain expertise is crucial.

The LLaMA-2-7B model, which was fine-tuned, outperformed the Basic Text Extraction Model, Llama2 with Prompt Engineering, and Gemini Pro with Prompt Engineering, thus proving the superiority of specialized training. It demonstrates the ability of language models that are capable not only of comprehending and summarizing content but also of delivering the summaries in a manner that shows depth of understanding, just as would an expert in the field.

In summary, the fine-tuned LLaMA-2-7B model adds a layer of precision and domain-specific expertise to the content creation process, complementing the foundational extraction capabilities of the Basic Text Extraction Model and the advanced summarization techniques employed by Llama2 and Gemini Pro with Prompt Engineering. Its inclusion in the evaluation highlights the advancements in language model training and the strides made towards creating AI that can generate text with an unprecedented level of relevance and insight.

To gauge the performance of these models, a comprehensive evaluation framework was devised, integrating both quantitative and qualitative assessments. Central to this framework was the solicitation of subjective feedback from human evaluators, whose perceptions and judgments were instrumental in elucidating the effectiveness of the models. Below table shows the details of Human Evaluation that we have performed.

S.No	Text Extraction			Llama-2 with Prompt Engineering			Gemini-Pro with Prompt Engineering			Fine Tuned llama-2 With Prompt Eng.		
	Easy	Medium	Hard	Easy	Medium	Hard	Easy	Medium	Hard	Easy	Medium	Hard
1	4	3	3	7	6	5	9	8	8	8	9	8
2	4	3	2	7	6	5	9	9	8	9	8	9
3	5	3	3	8	5	4	8	8	8	9	8	8
4	4	3	3	7	5	5	8	7	7	8	9	7
5	5	5	5	8	6	6	8	8	9	9	9	8
6	4	4	3	7	7	5	9	8	8	8	7	7
7	4	2	2	6	6	6	8	8	9	8	8	8
8	6	5	3	6	5	5	9	9	8	9	8	9
9	5	4	4	7	6	6	7	9	7	8	8	7
10	5	4	4	6	6	5	9	8	9	8	6	9
11	6	6	4	7	7	6	8	7	8	9	9	7
12	5	4	4	6	5	4	9	9	7	8	8	7
13	6	5	5	8	7	7	8	9	8	7	8	8
14	5	5	3	7	5	5	7	7	8	8	9	7
15	5	3	2	6	6	5	8	7	7	9	8	8
16	6	5	4	7	7	6	7	8	7	9	9	8
17	5	4	3	6	6	4	9	7	8	8	8	8
18	5	5	2	8	8	7	9	7	9	9	9	8
Avg	4.944	4.055	3.277	6.888	6.055	5.333	8.277	7.944	7.944	8.388	8.222	7.833

Table 1. : Manual Evaluation data of Text Generation Models

Upon analysis of the aggregated scores and expected accuracy metric, compelling insights emerged regarding the comparative performance of the evaluated models. While the basic text extraction model provided a rudimentary baseline, Llama2 with Prompt Engineering shows a better result, Gemini Pro with Prompt Engineering and fine tuned llama2 with prompt

engineering demonstrated significant advancements in content synthesis, we used fine tuned llama2 with prompt engineering as our final content generation model as it shows the best result among all models.

The overall result for each model is presented in the below table:-

Model	Text Extraction	Llama-2 with Prompt Engineering	Gemini-Pro with Prompt Engineering	Fine Tuned llama-2 With Prompt Eng.
Easy	4.944	6.888	8.277	8.388
Medium	4.055	6.055	7.944	8.222
Hard	3.277	5.333	7.944	7.833
Average	4.092	6.092	8.055	8.148

Table 2. : Mean Opinion Score for Text Generation Models

To provide stakeholders with a clear understanding of the evaluation outcomes, bar charts were generated, showcasing the average accuracy of each model. These visualizations served as powerful communicative tools, elucidating the nuanced differences in performance across the evaluated models.

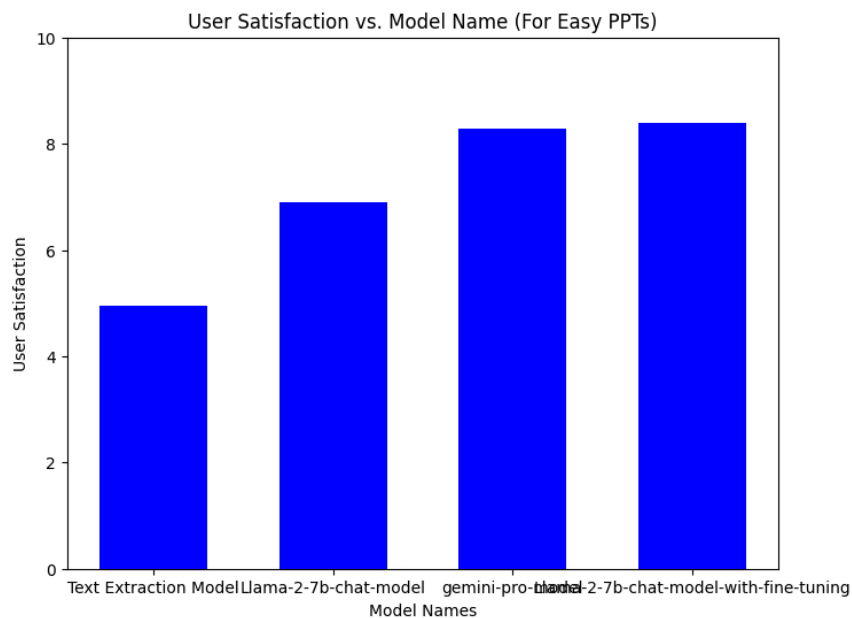


Fig. 10: User Satisfaction vs Model Name for easy level PPTs

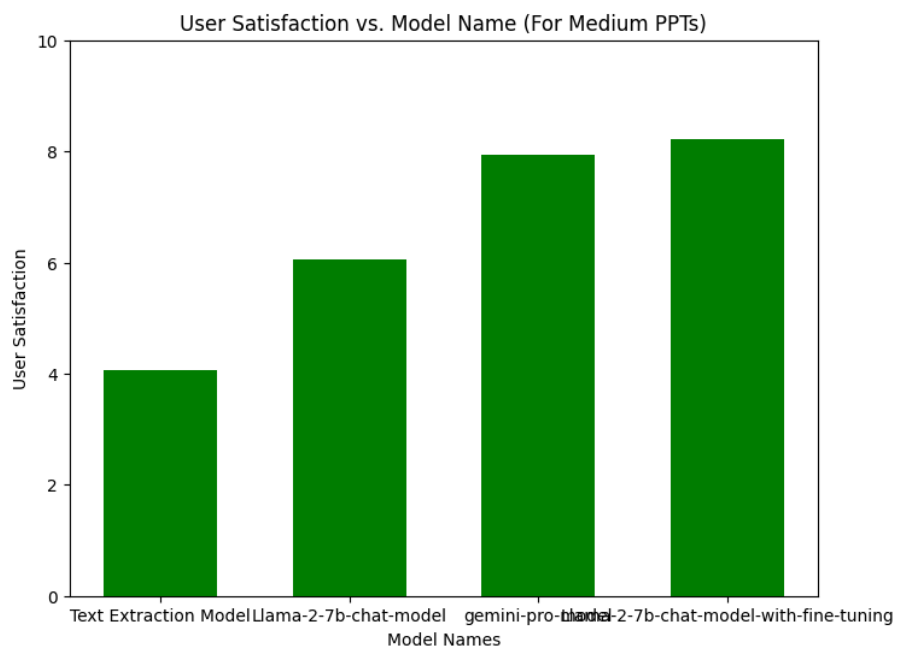


Fig. 11: User Satisfaction vs Model Name for medium PPTs

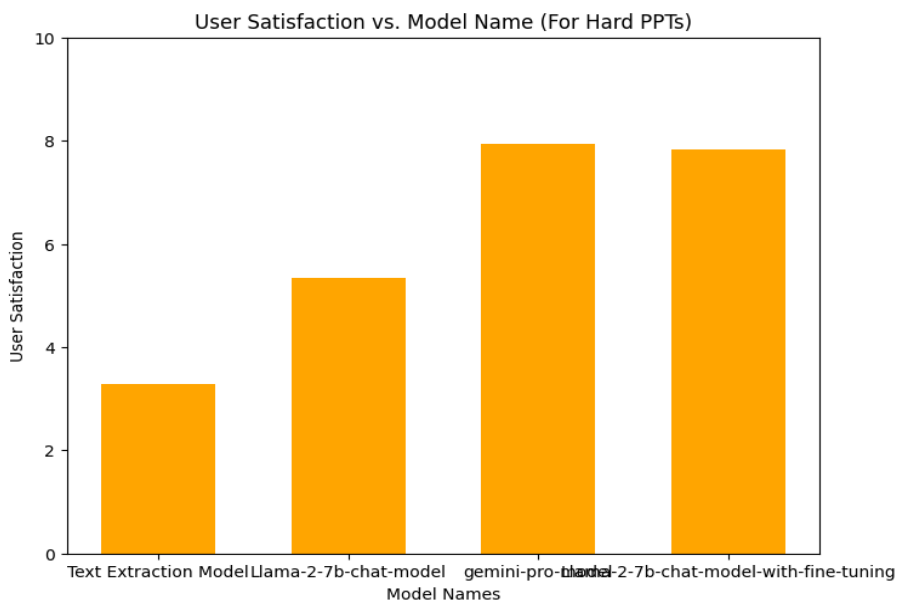


Fig.12 : User Satisfaction vs Model Name for hard PPTs

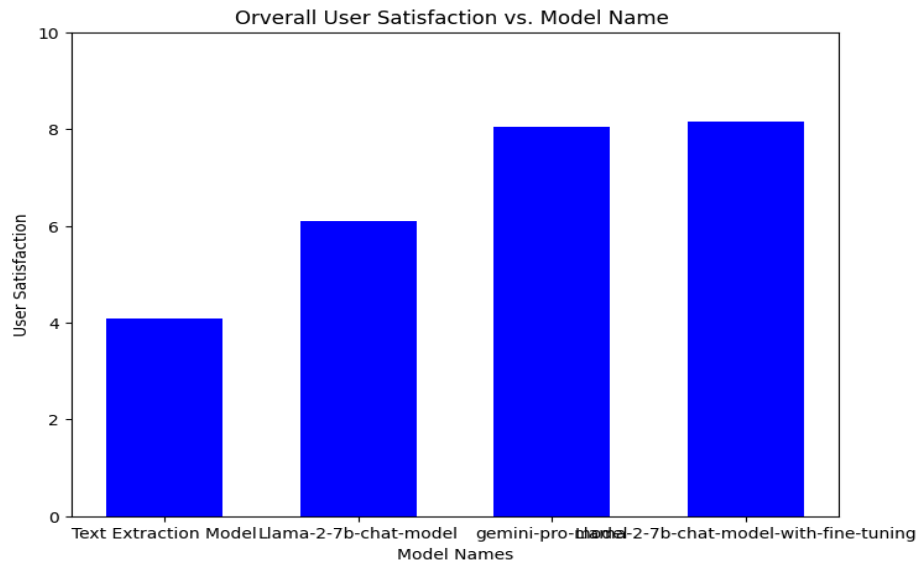


Fig.13 : Overall User Satisfaction vs Model Name

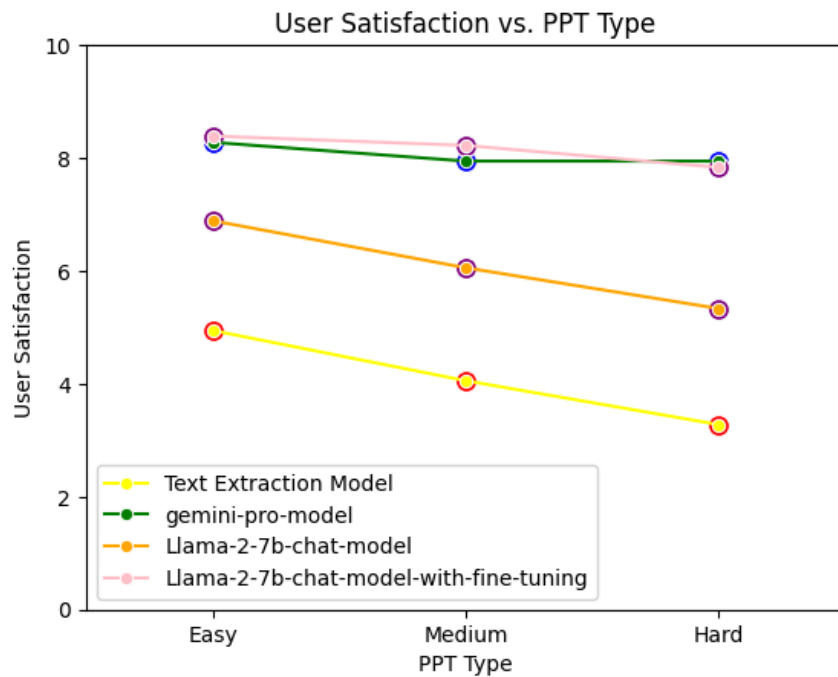


Fig.14 : Comparison of all four models on three categories of the PPTs

In the above figure we have shown the comparison of models based on giving different types of ppt. Fig 10 shows the compression of different models when input ppt is of essay type. Fig 11 shows when PPT is of medium type and Fig 12 is for hard type PPT. and fig 13 shows the overall comparison of all models.

5.3. Google Text-to-Speech Integration :

In the realm of Text-to-Speech (TTS) technology, the integration of Google Text-to-Speech API stands as a pivotal component in our project, providing an unparalleled auditory experience. Leveraging the power of Google's cutting-edge AI technologies, this API offers a robust set of features that elevate the quality of synthesized speech to near-human levels. The API boasts high-fidelity speech, a critical attribute derived from DeepMind's speech synthesis expertise. This results in voices that exhibit natural intonation, enhancing the overall user experience and making the educational content more engaging.

A WaveNet generates speech that sounds more natural than other text-to-speech systems. It synthesizes speech with more human-like emphasis and inflection on syllables, phonemes, and words. Here is a comparison of different text to speech models available in the market with their accuracy.

The y-axis values represent the Mean Opinion Score (MOS) for each voice. Test subjects ranked each voice on a scale of 1-5 according to how much it sounded like natural speech. For more information on MOS scores and WaveNet technology, see the [9] DeepMind WaveNet page.

The decision to integrate Google Text-to-Speech into our project was driven by its comprehensive feature set, wide language support, and the ability to create a unique brand voice. The API's reliability, scalability, and constant updates align with our commitment to providing a state-of-the-art educational experience. In conclusion, the Google Text-to-Speech API emerges as a preferred choice, elevating the auditory component of our AI-powered educational platform to new heights.

5.4. Lip-Sync Model Video Generation model :

Our evaluation method for the Lip-Sync Model involves real feedback from users and experts. To gauge the performance of these models, a comprehensive evaluation framework was devised, integrating both quantitative and qualitative assessments. Central to this framework was the solicitation of subjective feedback from human evaluators, whose perceptions and judgments were instrumental in elucidating the effectiveness of the models.

S.No.	Questions		
	How is the Video Quality?	How Good is Lip-Sync?	How is the Audio Quality?
1	8	8	9
2	8	7	8
3	8	7	9
4	7	7	7
5	8	6	7
6	9	8	7
7	10	7	9
8	8	8	7
9	6	7	7
10	8	7	7
11	9	8	6
12	8	8	8
13	9	8	7
14	7	7	6
15	8	6	6
16	7	7	7
17	8	8	7
18	9	7	7
Avg	8.0555	7.2777	7.2777

Table 3 . Manual Evaluation of Lip-Sync models

In feedback we have generated a final output using our model and given that video sample to all our experts and friends to give their feedback . We have asked three questions:

- 1 .How is the quality of video?
2. How good is Lip-Sync-Video?
3. How is the quality of audio?

On the basis of all these questions we have evaluated the lip-Sync model.

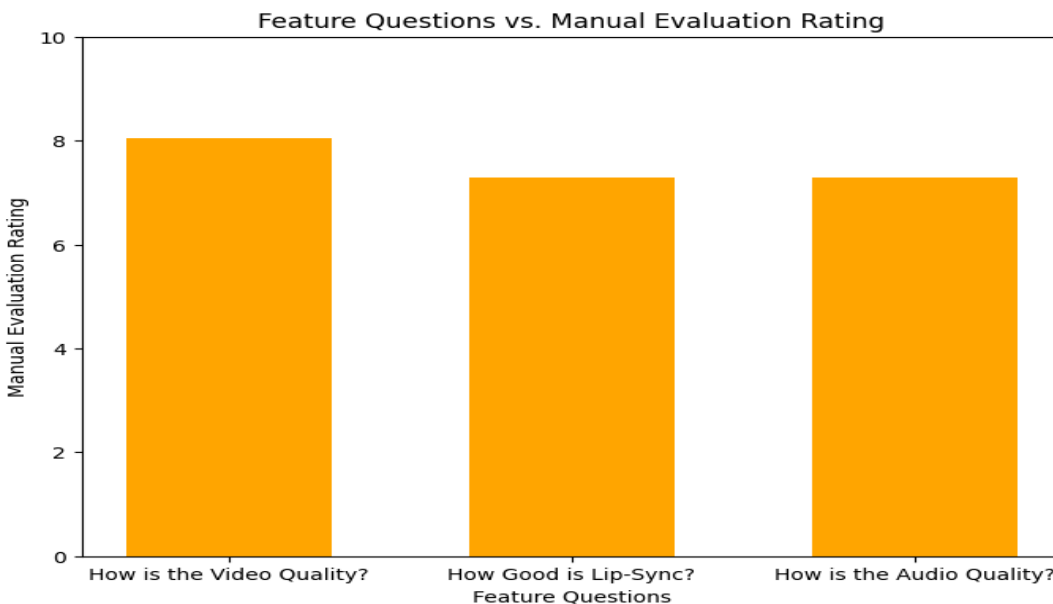


Fig 15 . Feature Questions Vs Manual Evaluation Rating

In the above figure we have shown the output of manual evaluation where we have collected opinions on our final generated video based on three question. Fig 14 shows the average rating given to the video based on each type of question.

CHAPTER 6

Web Application and Interface

To access the project we have developed a Web Application. The web application architecture is designed with a client-server model, with Flask serving as the backend server and React handling the frontend interactions. This approach ensures a clean separation of concerns, allowing for scalability, maintainability, and efficient data flow. The Flask server provides RESTful API endpoints, enabling communication between the backend and frontend components. The integration of our AI-powered PowerPoint presenter with a React frontend and Flask server proved to be instrumental in enhancing the user experience and facilitating seamless interaction with the system. Through the fusion of cutting-edge AI technologies with intuitive web interfaces, we achieved a dynamic and user-friendly platform for educational content delivery. The codebase contains more than 80,000+ lines of code.

Technology Used	Description
HTML, CSS	HTML and CSS are utilized for designing the UI/UX of our project, ensuring a visually appealing and user-friendly interface.
JavaScript	JavaScript is used as a client-side language to connect the AI model with the website, enabling dynamic interactions and real-time feedback.
React	React is a JavaScript library for building user interfaces, used for the frontend of our web application. It provides an interactive and intuitive user interface through its component-based architecture.

SCSS	SCSS (Sassy CSS) is a preprocessor scripting language that is interpreted or compiled into Cascading Style Sheets (CSS). It enhances the readability and maintainability of CSS code by providing features such as variables, nesting, and mixins.
Flask	Flask is a Python web framework serving as the backend for our web application. It handles key functionalities such as authentication, data processing, and communication with external APIs.
SQLAlchemy	SQLAlchemy is used for backend operations, providing an Object-Relational Mapping (ORM) for interacting with the database.

TABLE 4: Tech stack of Web Integration



Fig.16 : Frontend and Backend codebase structure

6.1 User-login system

The robust authentication system employed in our project serves as a critical component ensuring secure access to the educational platform. Utilizing token-based authentication, we establish a reliable mechanism for verifying the identity of users and granting access to authorized resources.

Implementing JWT (JSON Web Token) based authentication adds a layer of security to our educational platform, ensuring that only authorized users can access protected resources. JWT works by securely transmitting information between parties as a JSON object, which is then signed using a secret key. This signed token can be verified and trusted because it's digitally signed.

Upon successful login, the server issues a JWT containing user information and a timestamp. This token is then sent to the client, typically stored in local storage or cookies, and included in subsequent requests to the server. The server validates the token's signature and expiration time, granting access to protected endpoints if the token is valid.

JWT-based authentication offers several advantages:

- **Statelessness:** Since JWTs contain all necessary information, servers don't need to store session data, improving scalability and reducing server load.
- **Security:** Tokens are digitally signed, ensuring that they haven't been tampered with. Additionally, sensitive information such as passwords can be omitted from tokens, enhancing security.
- **Cross-Domain Compatibility:** JWTs can be easily transmitted across different domains, facilitating authentication in distributed systems.

In our educational platform, JWT-based authentication enhances security and ensures that only authenticated users can access sensitive resources. By securely transmitting and verifying tokens, we uphold the integrity of user data and protect against unauthorized access.

6.2 User Interface

The user interface of our educational platform is meticulously designed to provide a seamless and intuitive experience for learners of all backgrounds. With a clean and modern aesthetic, the interface employs a soothing color palette and intuitive layout, guiding users effortlessly through their educational journey. Interactive elements are thoughtfully placed to enhance engagement, while clear navigation ensures easy access to all features and resources. From uploading PPT

files to interacting with AI-generated content, every aspect of the interface is optimized for user friendliness, fostering a sense of comfort and ease as users explore and engage with the platform's rich educational offerings.

6.3 Database Integration

Our Flask application harnesses the power of SQLite as the backend database, providing a lightweight yet powerful solution for managing application data. SQLite's serverless architecture eliminates the need for complex setup and configuration, simplifying deployment and maintenance while offering robust transactional support and ACID compliance.

Our Flask application leverages the SQLAlchemy ORM to interact with the SQLite database, abstracting away the complexities of SQL queries and providing a Pythonic interface for database operations. With SQLAlchemy handling database management tasks, our educational platform can focus on delivering rich content and engaging experiences, confident in the reliability and efficiency of our data management solution.

6.4 Local network deployment

Deploying our educational platform on a local network offers the flexibility for seamless access from any device within the same network. By configuring the backend server to allow connections from other devices, coupled with enabling CORS (Cross-Origin Resource Sharing) in the frontend, we facilitate unrestricted access to the platform's features and content. This setup ensures that users within the network can access the platform by simply navigating to the host machine's IP address and port where the server is running, fostering collaboration and accessibility among users on various devices within the local network. Whether it's accessing educational resources, interacting with AI-powered features, or collaborating on learning activities, our platform empowers users to engage and learn from anywhere within the network with ease and convenience.

CHAPTER 7

Conclusion and Future Scope

In conclusion, our innovative AI-powered educational platform, integrating advanced technologies such as text summarization, script generation, and Google Text-to-Speech API, presents a transformative solution to the limitations of traditional PowerPoint-based learning. By prioritizing personalization, accessibility, and user engagement, we have created a dynamic tool that bridges educational gaps, offering a cost-effective, 24/7 accessible teacher-like experience. The successful web integration with Flask and React ensures a seamless and interactive interface. The choice of Google Text-to-Speech API enriches the auditory component with high-fidelity, near-human voices, enhancing the overall educational journey. Our project marks a significant stride towards inclusive and adaptive learning environments.

AI Powered Voice-Guided Presentation: Revolutionizing PPT Explanations can benefit various applications in fields like online education, news anchoring using AI robots in custom voice, reduce teachers workload etc. In the future we can think about how we can integrate lip syncing where we are able to deliver a ppt explanation in any custom voice with video in which that particular person's video will be played and it will seem like the person's real video. And in future we will work on how we can increase preciseness and accuracy of our ppt explanation by trying different models and doing hypertuning of existing model

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Appendix I

I.1 Code

The Complete source code of our project is available on Github at the following Link

Github link : <https://github.com/shibam-debnath/EduVista>

I.2 Installation Instruction

I.2.1 Prerequisites :

- Node.js installed on your system
- Python installed on your system

I.2.2 Installation steps :

- Clone the Repository:
 - git clone <https://github.com/shibam-debnath/EduVista>
 - cd your-project
- Setup Frontend:
 - cd frontend
- Install dependencies:
 - npm install
- Run the development server:
 - npm run dev
- Your React app should now be running at <http://localhost:3000>.
- Setup Backend:

- `cd ../backend`
- Create a virtual environment (optional but recommended):
 - `python -m venv venv`
- Activate the virtual environment:
 - On Windows:
 - `venv\Scripts\activate`
 - On macOS and Linux:
 - `source venv/bin/activate`
- Install Flask and other dependencies:
 - `pip install -r requirements.txt`
- Set up environment variables (if required) in a `.env` file.
- Run the Flask development server:
 - `python app.py`
- Your Flask backend should now be running at `http://localhost:5500`.
- Configure Frontend to Communicate with Backend:
 - In your React app's code, ensure that API requests are made to `http://localhost:5000` (or whichever port your Flask server is running on) for backend communication.

I.2.3 Usage

- Open the directory in your code editor and run frontend and backend.
- To run frontend execute command `npm run dev`
- To run the backend first activate the virtual environment and then run `python app.py`.
- The frontend will be active on port 5173 and flask server will be active on port 5500.

Appendix II

II.1 User Interface

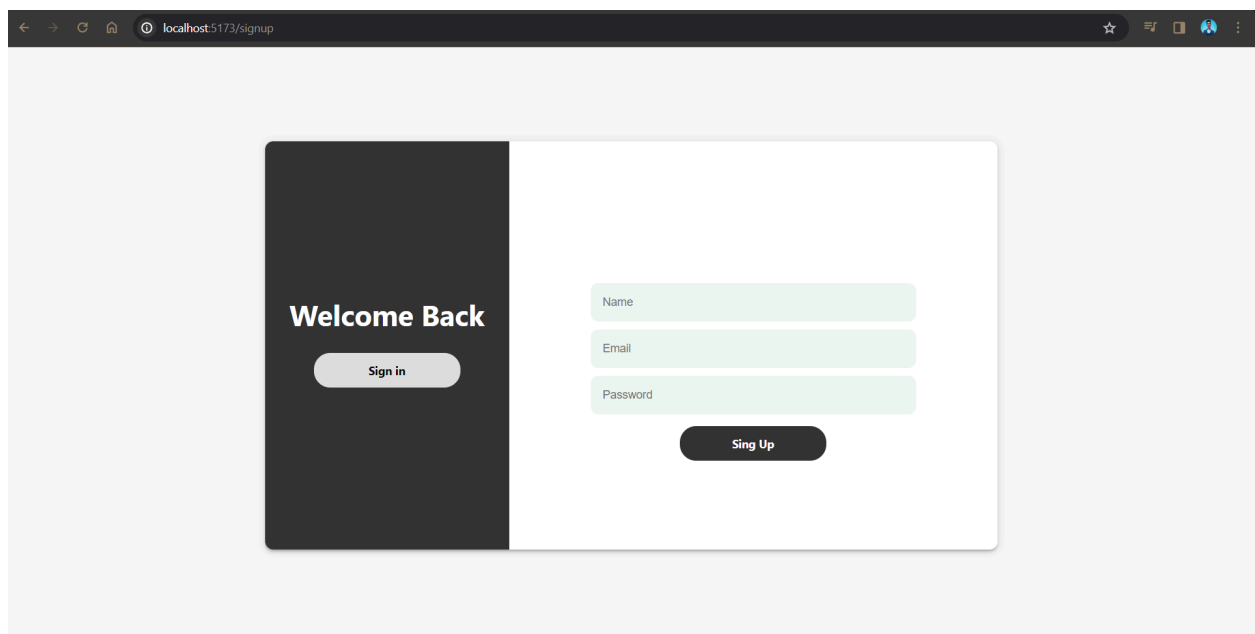


Fig.II.1.1 : Sign in And Sign Up Page

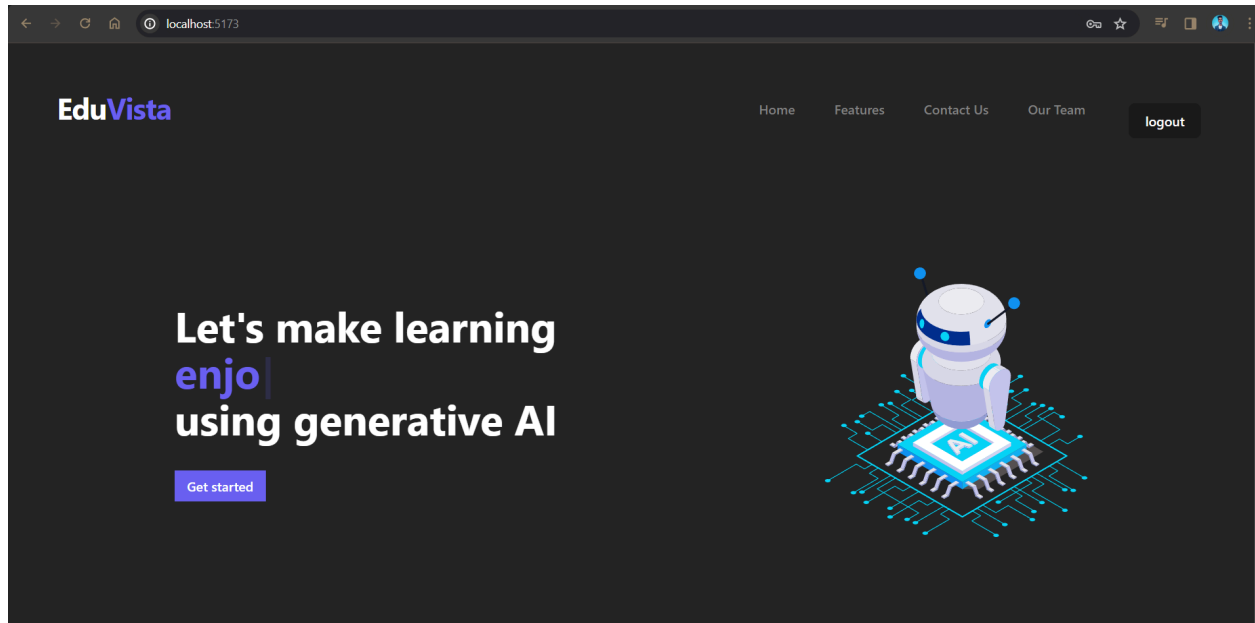


Fig.II.1.2 : Dashboard of Website

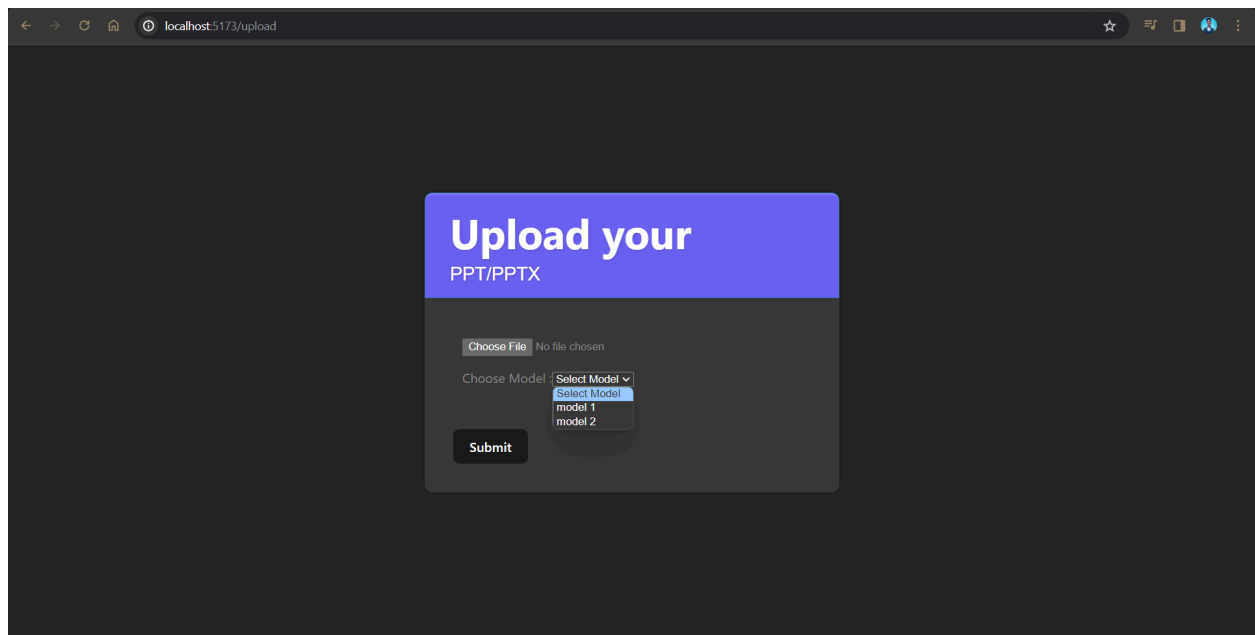


Fig.II.1.3 : Upload ppt and choose model button

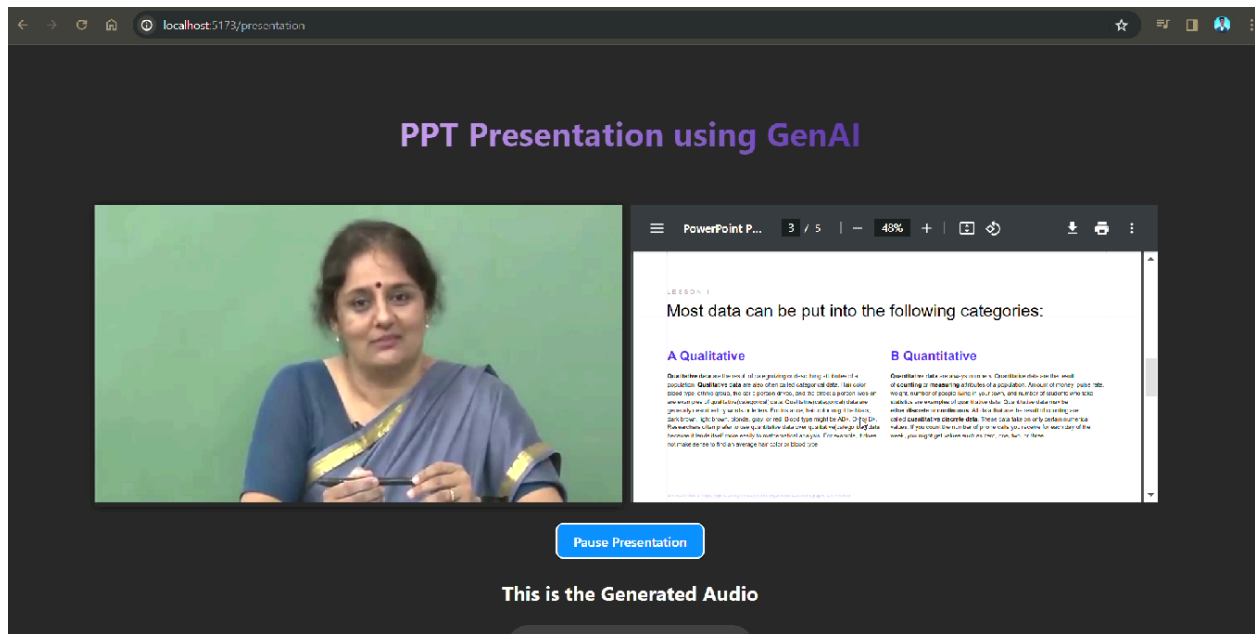


Fig II.1.4 : Lip-Sync video and ppt display on dashboard